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$$\begin{aligned} & \int \frac{5x+2}{x-3} dx \\ &= \int \frac{5x-15+17}{x-3} dx = \int \frac{5(x-3)}{x-3} dx + \int \frac{17}{x-3} dx \\ &= 5 \int \frac{x-3}{x-3} dx + 17 \int \frac{1}{x-3} dx \\ & z = x-3, dz = 1dx \\ &= 5 \int 1 dx + 17 \int \frac{dz}{z} = 5x + 17 \ln |z| + c \\ &= 5x + 17 \ln |x-3| + c \end{aligned}$$

References